

Project Documentation

Project Title : Smart Library Request Workflow in ServiceNow

1.Introduction

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1.Identification Phase

The primary goal of this phase is to modernize traditional library management by replacing manual, error-prone processes with an automated system.

Problem Statement

Traditional systems face significant operational hurdles:

Lack of Real-time Data: Staff must manually check registers to verify if a book is available, often leading to students requesting already-issued items.

Inefficient Approvals: Manual review processes are slow and prone to human error.

Limited Visibility: Administrators struggle to track borrowing patterns or popular titles without digital analytics.

Data Risks: Physical record-keeping increases the risk of data inconsistency, duplication, and loss.

2. Requirement Analysis

This phase defines the roles, data structures, and functional capabilities necessary for the system.

User Roles

The system utilizes Role-Based Access Control (RBAC) to maintain security:

Student: Can search for books, check availability, submit borrow requests, and track their history.

Librarian: Manages the catalog (adding/updating books), reviews requests, and issues books.

Database Tables

The system relies on two primary tables to organize information:

Book Table (u_book): Stores Title, Author, Category, and Availability Status.

Borrow Request Table (u_borrow_request): Tracks student requests and transaction details.

3. Project Planning Phase

Planning ensures a systematic implementation by defining milestones and a roadmap.

Key Milestones

Role Definition: Configuring permissions for Students and Librarians.

Table Structure Design: Designing the schema for the book and request tables.

Workflow Creation: Developing the automated logic for borrowing and approvals.

4. Project Development Phase

This phase focuses on the technical automation and communication features of the application.

Workflow Automation

Using ServiceNow Flow Designer, the system automates several manual steps:

Upon librarian approval, the book status in the u_book table automatically changes to "Issued".

This prevents other students from requesting the same book until it is returned.

Business Rules & Notifications

Real-time Updates: A Business Rule triggers an automated email to the student whenever a request is approved or rejected.

Reference Qualifiers: These filters ensure students can only see and select books currently marked as "Available" on the request form.

5. Project Design Phase

The design phase focuses on the User Interface (UI) and data integrity.

UI Policies

UI Policies are implemented to make forms dynamic and user-friendly:

Conditional Mandatory Fields: For example, the "Return Date" field only becomes mandatory if the book status is set to "Issued".

Benefits: This simplifies the user experience and prevents the entry of incomplete or incorrect data.

6. Project Documentation Phase

Documentation ensures the system is maintainable and reproducible.

Documentation Components

Technical Blueprint: Details the internal structure, including variables like Borrow Date, Return Date, and Request Status.

Setup Manual: A step-by-step guide for recreating the system in a Personal Developer Instance (PDI), covering table creation, role configuration, and workflow testing.

7. Project Demonstration Phase

The final phase showcases the system's real-time functionality from a user perspective.

The Borrowing Lifecycle

The demonstration highlights the following automated stages:

Requested: Student submits the form via the Service Portal.

Approved: Librarian reviews and approves the request in the dashboard.

Issued: System automatically updates book status to "Issued".

Returned: Librarian updates the status upon return, and the book becomes "Available" again.

Next Step: Would you like me to generate a summary table comparing the manual traditional process versus the new ServiceNow automated process?

